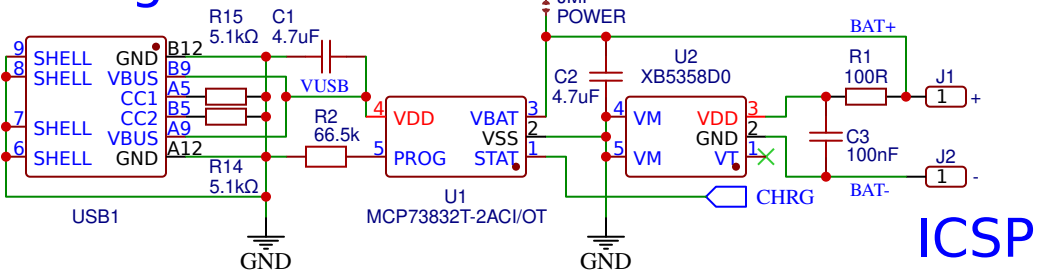
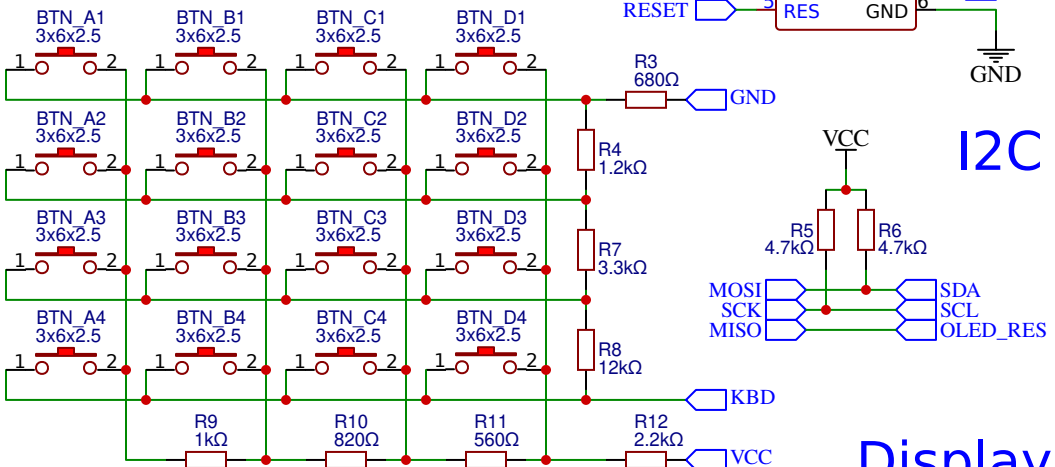


[illegible]

The diagram illustrates a 4x4 matrix keyboard circuit. The keyboard consists of 16 buttons, each labeled with a 3x6x2.5 specification. The buttons are arranged in a 4x4 grid, with columns labeled A1 through D4 and rows labeled B1 through B4. The keyboard is connected to an ICS9650B I2C-to-USB bridge via a 4-wire interface: MISO, SCK, RESET, and RES. The bridge is connected to a USB port. The keyboard is also connected to an SSD1306 OLED display via an I2C interface. The I2C interface uses two 4.7kΩ pull-up resistors (R5 and R6) connected to VCC. The I2C lines are labeled SDA, SCL, and OLED_RES. The keyboard is powered by a 5V supply (VCC) and ground (GND). The display is powered by a 5V supply (VCC) and ground (GND). The display is connected to the I2C interface via a 2.2kΩ resistor (R12) and a 1kΩ resistor (R9). The display is also connected to the I2C interface via a 1.2kΩ resistor (R4) and a 3.3kΩ resistor (R7). The display is connected to the I2C interface via a 12kΩ resistor (R8) and a 560Ω resistor (R11). The display is connected to the I2C interface via a 820Ω resistor (R10) and a 1kΩ resistor (R9).



The schematic diagram illustrates the hardware setup for the 'Clock' project. It features three main components: an RTC (DS3231MZ+), an MCU (ATTINY85-20SU), and an OLED display (SSD1306 128X32).

RTC (DS3231MZ+): The RTC is connected to the MCU via I2C. The SCL pin (pin 8) is connected to the MCU's PB5 pin (pin 1). The SDA pin (pin 7) is connected to the MCU's PB3 pin (pin 2). The RST# pin (pin 4) is connected to the MCU's PB4 pin (pin 3). The VBAT pin (pin 6) is connected to the MCU's PB1 pin (pin 5). The GND pin (pin 5) is connected to the MCU's GND pin (pin 4). The VCC pin (pin 2) is connected to the MCU's VCC pin (pin 8).

MCU (ATTINY85-20SU): The MCU is connected to the OLED display via I2C. The SCL pin (pin 8) is connected to the display's SCL pin (pin 10). The SDA pin (pin 7) is connected to the display's SDA pin (pin 11). The RST# pin (pin 4) is connected to the display's IREF pin (pin 13). The VBAT pin (pin 6) is connected to the display's VCOMH pin (pin 14). The GND pin (pin 4) is connected to the display's GND pin (pin 12). The VCC pin (pin 8) is connected to the display's VCC pin (pin 14).

OLED Display (SSD1306 128X32): The display is connected to the MCU via I2C. The SCL pin (pin 10) is connected to the MCU's SCL pin (pin 8). The SDA pin (pin 11) is connected to the MCU's SDA pin (pin 7). The RST# pin (pin 13) is connected to the MCU's RST# pin (pin 4). The VCOMH pin (pin 14) is connected to the MCU's VBAT pin (pin 6). The GND pin (pin 12) is connected to the MCU's GND pin (pin 4). The VCC pin (pin 14) is connected to the MCU's VCC pin (pin 8).

Power and Timing: The RTC is powered by a 32kHz crystal (pin 1) and a 100nF capacitor (C8). The MCU is powered by a 100nF capacitor (C6). The OLED display is powered by a 1000V 66.5k resistor and a 15 mA current source. The display is also connected to a 100nF capacitor (C7) and a 2.2uF capacitor (C10).

The schematic diagram illustrates the hardware components and their interconnections for the 'The Time Machine' project. The components include an RTC (DS3231MZ+), an MCU (ATTINY85-20SU), and an SSD1306 128X32 OLED display.

RTC (DS3231MZ+) Connections:

- Pin 1 (32kHz): Connected to a 32kHz crystal.
- Pin 3 (INT#/SQW): Connected to a 100nF capacitor.
- Pin 5 (GND): Connected to ground.
- Pin 8 (SCL): Connected to the SCL pin of the MCU.
- Pin 7 (SDA): Connected to the SDA pin of the MCU.
- Pin 4 (RST#): Connected to a 100nF capacitor.
- Pin 6 (VBAT): Connected to a 100nF capacitor.
- Pin 2 (VCC): Connected to the VCC pin of the MCU.

MCU (ATTINY85-20SU) Connections:

- Pin 1 (RESET): Connected to a 100nF capacitor.
- Pin 2 (CHRG): Connected to a 100nF capacitor.
- Pin 3 (KBD): Connected to a 100nF capacitor.
- Pin 4 (GND): Connected to ground.
- Pin 5 (PB0): Connected to the MOSI pin of the OLED display.
- Pin 6 (PB1): Connected to the MISO pin of the OLED display.
- Pin 7 (PB2): Connected to the SCK pin of the OLED display.
- Pin 8 (VCC): Connected to the VCC pin of the OLED display.

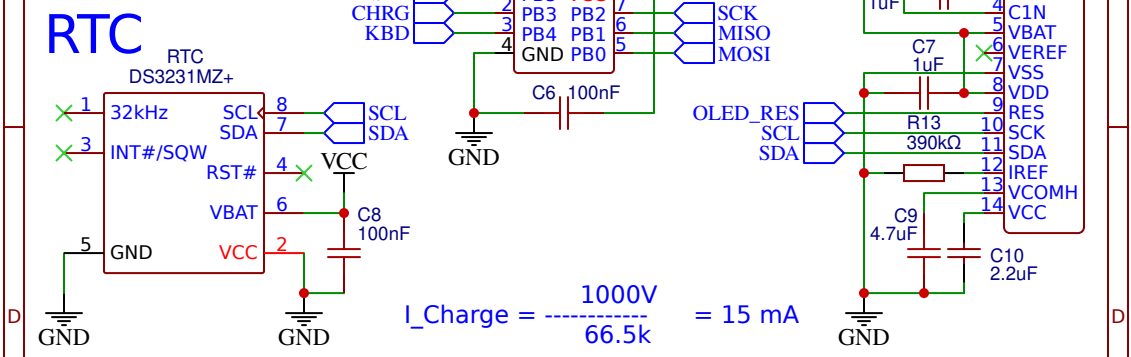
SSD1306 128X32 Connections:

- Pin 1 (C2P): Connected to a 100nF capacitor.
- Pin 2 (C2N): Connected to a 100nF capacitor.
- Pin 3 (C1P): Connected to a 100nF capacitor.
- Pin 4 (C1N): Connected to a 100nF capacitor.
- Pin 5 (VBAT): Connected to a 100nF capacitor.
- Pin 6 (VEREF): Connected to a 100nF capacitor.
- Pin 7 (VSS): Connected to ground.
- Pin 8 (VDD): Connected to the VCC pin of the MCU.
- Pin 9 (RES): Connected to a 390k resistor.
- Pin 10 (SCK): Connected to the SCK pin of the MCU.
- Pin 11 (SDA): Connected to the SDA pin of the MCU.
- Pin 12 (IREF): Connected to a 100nF capacitor.
- Pin 13 (VCOMH): Connected to a 100nF capacitor.
- Pin 14 (VCC): Connected to the VCC pin of the MCU.

Power Supply and Grounding:

- The VCC pin of the MCU is connected to the VCC pin of the OLED display.
- The GND pin of the MCU is connected to the GND pin of the OLED display.
- The VCC pin of the RTC is connected to the VCC pin of the MCU.
- The GND pin of the RTC is connected to the GND pin of the MCU.

Current Calculation:

$$I_{\text{Charge}} = \frac{1000V}{66.5k} = 15 \text{ mA}$$


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TITLE: Business Card RPN Calculator with Watch		REV: 1.5
	Company: Olexandrenko	Sheet: 1/1
	Date: 2026-09-10 Drawn By: Yevgeniy	